

within a certain level there are also decorations for instructions (I) and data (D). In Zen, the L1-I instruction cache is not shared and both, the L1-D and L1-I have lower latency than before.

The Zen architecture has more lower level cache closer to the core. Low-level caches are exclusive to cores and faster as well. So it helps increase the performance per core i.e. single threaded performance.

SSH - SIMULTANEOUS MULTI-THREADING

With Bulldozer AMD had used Clustered Multi-Threading (CMT) to enable multiple threads within the same core. This way, one CMT module was thought off as two Cores. Unfortunately, that didn't work out properly as one CMT module shared a Floating Point pipeline. So it was more like each CMT being 1.5 Cores. With Zen, AMD has adopted Simultaneous Multi-Threading (SMT) which is also used by Intel. In SMT, each core has two virtual cores wherein workload gets distributed between them. Theoretically, CMT should be more efficient but it was quite the opposite when it came to real world applications.

INTEL'S SKYLAKE CORE

Intel's Skylake architecture was introduced in 2015 and has been at the core of all "Lake" processors since then. The major change in the architecture was the removal of the (Fully Integrated Voltage Regulator) from the CPU. FIVR allowed the CPU to have a more granular control on the voltage levels. Unfortunately, this also generated a lot of heat which made it difficult for the processors to overclock. It's rumoured that an improved version of FIVR should be coming out later this year with Ice Lake processors.

The other major changes included new instruction sets in the form of the Intel MPX (Memory Protection Extensions) and SGX (Software Guard Extensions) which are related to security. The Interconnect between the processor and the chipset (also known as PCH) also received a shot in the arm with the introduction of DMI 3.0. This increased the speed to 8 GT/s and

JARGON BUSTER

BGA

A ball grid array uses an array of solder balls to mount an integrated circuit. The delicate process is usually automated.

Caching

A cache is a temporary memory closer to the processor core, speeding up operations by reducing the time required to access the main memory. The most frequently used data is stored in the cache.

CAS

The column address select is a signal sent from the CPU or GPU to the RAM containing information on which column of the memory has the requested information.

Clock Speed

The clock speed is a rough representation of how fast a processor is, but these are not strictly comparable as different processor architectures can require varying number of clock cycles to complete the same instruction.

Context Switch

The context of the process being run is saved as a set of register values and other associated data. The OS kernel uses this context to handover control of the CPU from a running process to another one that is ready to go.

Control Unit

This is the component of the CPU that informs the memory, the ALU and the I/O devices on how to respond to instructions from a program. If the CPU is a circus, then the CU is the ringmaster

requires more physical connections between the PCH and the CPU.

Most of the other feature introductions with the Skylake platform have to do with the PCH and not so much with the CPU itself. Sandy Bridge was in fact a much more significant upgrade in Intel's history. Several bottlenecks such as the instruction fetch and pre-decoding, limited register read ports, doubling of the memory read ports and an improved branch prediction logic were the significant ones. On the part of the ISA, the AVX instruction set was a massive improvement. The list of changes for the Zen core was much more because it was a monumental shift from their usual strategies.

OF CLOCK SPEEDS AND BENCHMARKS

You've no doubt seen a massive list of processor benchmarks whenever each new CPU is introduced by either Intel or AMD. Most benchmarks simply get down to what benchmarks score more on which processor. However, the real world impact of each benchmarks are rarely discussed and understood by even fewer folks. With CPUs, the benchmarks can easily be broken down into a few categories - performance, energy efficiency, performance/watt and stability. Let's take a look at the different benchmarks and what they indicate.

1. Compression/Decompression - This indicates how fast you'll be able to zip and unzip files.
2. Encode/Decode - These are similar to the encryption and compression benchmarks and tells you how fast the CPUs will be able to perform tasks such as video encoding from one format to another. Whether software that use DRM will work smoothly on your system.
3. Rendering - Rendering tasks are highly parallelised tasks which can be commonly seen in video editing and transcoding software. Even software which are used for 3D modelling will benefit from better rendering numbers.
4. Encryption - Several programs implement encryption to ensure that the data they work on is not